

Online Parking Slot Booking System

A Project Synopsis Report Submitted To



SAGE UNIVERSITY, INDORE

Towards Partial Fulfillment for the Award of

**Bachelor of Technology
in
Computer Science & Engineering**

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1. Problem Statement

In urban areas, finding an available parking space is a major challenge. Drivers often spend a significant amount of time searching for parking, leading to traffic congestion, fuel wastage, and frustration. Traditional parking systems do not provide real-time information about slot availability, making the parking process inefficient.

The proposed **Parking Slot Booking System** aims to provide a digital platform where users can search, reserve, and pay for parking slots in advance, ensuring a smooth and convenient parking experience.

2. Description of the Problem

With the rapid growth of vehicles in cities, parking management has become increasingly difficult. Many parking facilities operate manually, causing issues such as:

- Lack of real-time parking information.
- Time-consuming search for vacant slots.
- Traffic congestion near parking areas.
- Inefficient utilization of parking spaces.
- Difficulty in managing reservations and payments.

The system will solve these problems by providing an online booking platform with real-time slot availability and digital payment facilities.

3. Objectives and Scope of the Project

Objective

- To provide a platform for searching available parking slots.
- To allow users to reserve parking spaces in advance.
- To enable secure online payment for parking reservations.
- To display real-time parking slot availability.
- To provide booking confirmation through QR code or PIN.
- To help parking owners manage slots and reservations efficiently.

Scope

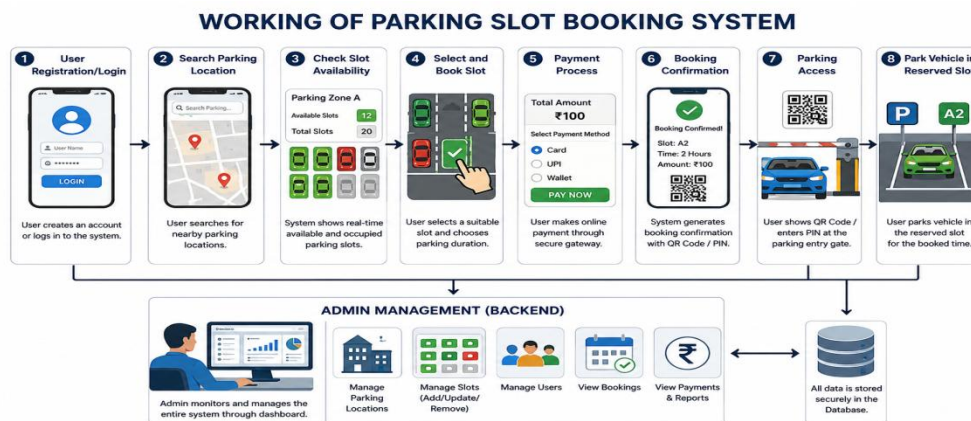
The project can be used in:

- Shopping malls
- Airports
- Railway stations
- Hospitals
- Colleges and universities
- Smart city parking systems

The system benefits both vehicle owners and parking facility operators by improving efficiency and reducing congestion.

4. Methodology

The development process will follow these steps:



Step 1: User Registration/Login

Step 2: Search Parking Area

Step 3: View Availability

Step 4: Slot Booking

Step 5: Online Payment

Step 6: Confirmation

Step 7: Parking Management

Parking owners manage slots, pricing, and reservations through an admin dashboard.

5. Hardware and Software Requirements

Hardware Requirements

- Computer/Laptop
- Internet Connection
- Smartphone (for users)
- Server System

Software Requirements

Frontend

- HTML
- CSS
- JavaScript

Backend

- Node.js / PHP /

Database

- MySQL / MongoDB

Additional Tools

- Google Maps API
- Payment Gateway Integration
- VS Code
- Git & GitHub

6. Expected Deliverables

- Complete Parking Slot Booking Website/Application.
- Parking facility listing with pricing and location details.
- Real-time slot availability system.
- Online reservation functionality.
- Booking history dashboard.
- QR Code/PIN-based entry system.

7. Future Work

The system can be enhanced with:

- Dynamic pricing during peak hours.
- EV charging slot booking.
- Monthly and weekly subscription passes.

- License plate recognition.
- Remote booking extension.
- Analytics dashboard for facility owners.
- Reviews and ratings system.
- Mobile application support.

Early exit refunds and overstay auto-charging

8. Project Schedule (Gantt Chart)

Activity	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Requirement Analysis	✓					
System Design	✓	✓				
Database Design		✓	✓			
Frontend Development			✓	✓		
Backend Development			✓	✓	✓	
Testing					✓	
Documentation					✓	✓
Final Deployment						✓

9. Conclusion

The **Parking Slot Booking System** is a smart and efficient solution for modern parking management. It reduces the time spent searching for parking spaces, minimizes traffic congestion, and provides a convenient booking experience through online reservations and payments. The system also helps parking facility owners manage resources effectively, making it a valuable contribution toward smart city infrastructure

10. References

1. HTML, CSS, JavaScript Documentation .
2. MySQL Documentation.
3. Google Maps API Documentation.
4. Node.js Documentation.
5. Research papers on Smart Parking Systems and IoT-based Parking Management.